3. Problems in Mathematical Analysis I — Kaczor & Nowak

- Strong problem-solving focus
- Many exercises naturally lead to:
  - constructing examples.
  - breaking false statements
- Widely recommended as a $\boxed{\text{core problem book in analysis.}}$

4. Real Analysis via Sequences and Series — Brunt, Little, Teo

- Develops analysis through sequences
- Includes $\boxed{\text{carefully chosen examples and counterexamples}}$.

5. Principles of Mathematical Analysis — Walter Rudin

- Very rigorous
- Dense
- Forces you to: 1) think precisely 2) understand necessity of assumptions

Problems

1. Find a formula for the nth term from the first several terms of the sequence

(a) $\frac{1}{11}, \frac{1}{21}, \frac{1}{31}, \frac{1}{41}, \frac{1}{51}, \ldots\ldots \rightarrow y_n = \frac{1}{10n+1}$ ✓

(b) $1, \frac{1}{4}, \sqrt{3}, \frac{1}{16}, \sqrt{5}, \frac{1}{36}, \sqrt{7}, \ldots\ldots \rightarrow y_n = \sqrt{n}$ if n is odd, $y_n = \frac{1}{n^2}$ if n is ev ✓

(c) $1, -\left(\frac{1}{2}\right)^2, \left(\frac{1}{2\cdot 3}\right)^3, -\left(\frac{1}{2\cdot 3\cdot 4}\right)^4, \ldots\ldots \rightarrow y_n = (-1)^{n+1} \cdot \left(\frac{1}{n!}\right)^n$ ✓

(d) $\frac{3}{4}, -\left(\frac{6}{7}\right)^2, \left(\frac{9}{10}\right)^3, -\left(\frac{12}{13}\right)^4, \left(\frac{15}{16}\right)^5, \ldots\ldots \rightarrow y_n = (-1)^{n+1} \cdot \left(\frac{3n}{3n+1}\right)^n$ ✓

2. Find the least term of each sequence:

(a) $y_n = n^2 - 5n + 1$ $\quad y_{n+1} = (n+1)^2 - 5(n+1) + 1$ $\quad y_{n+1} - y_n = \boxed{2n-4}$

$2n-4 > 0 \Rightarrow n > 2$. For $n > 2$, the difference between consecutive terms is always positive. $n=1$, $2n-4=-2$ $\quad n=2$, $2n-4=0$. $\therefore$ The second term must be minimum. $\boxed{y_2 = 2^2 - 5\times 2 + 1 = -5} = y_3$ ✓